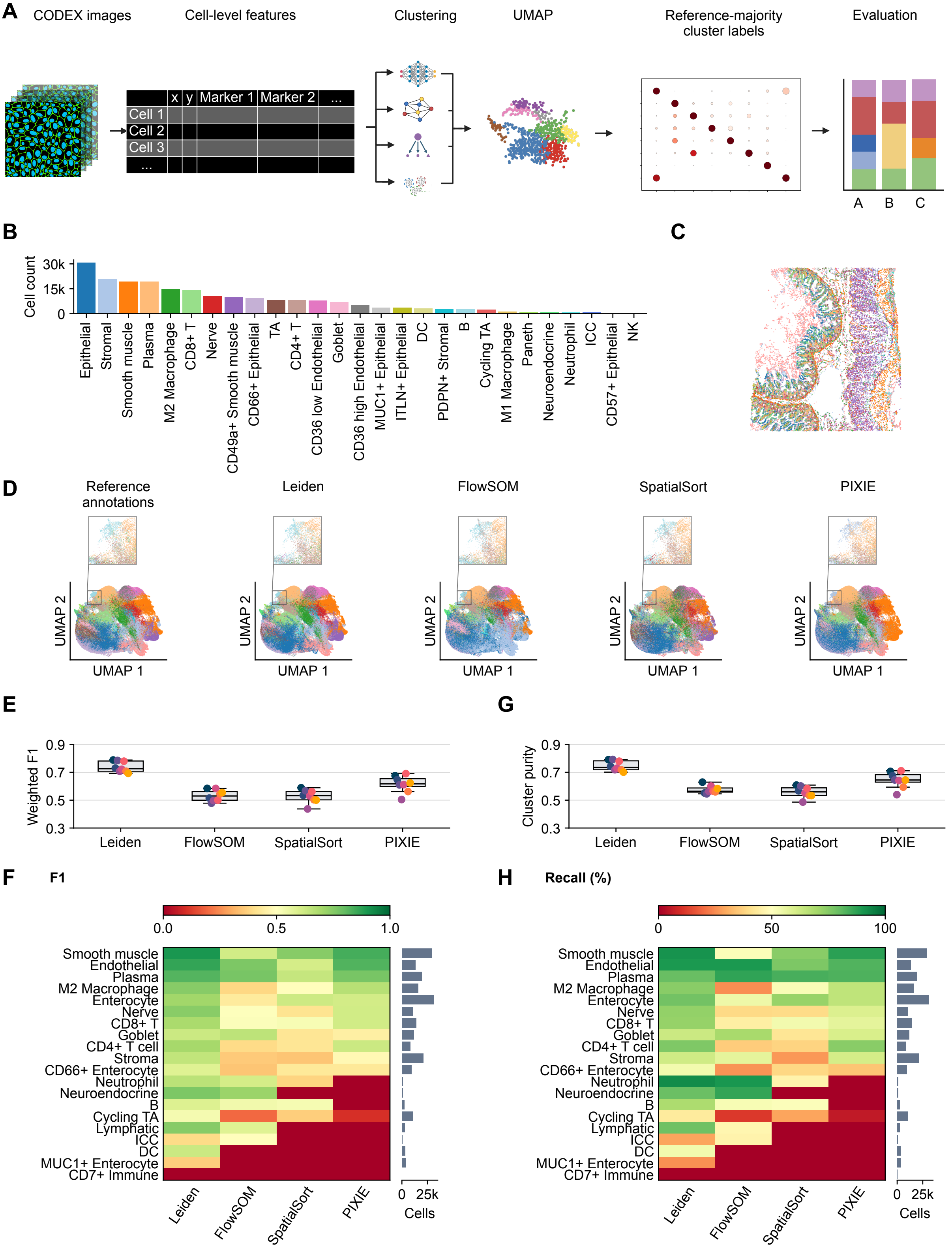
A
CODEX images
Cell-level features
x
y
Marker 1
Marker 2
...
Cell 1
Cell 2
Cell 3
...
Clustering
UMAP
Reference-majority cluster labels
Evaluation
A
B
C
B
Cell count
30k
15k
0
Epithelial
Stromal
Smooth muscle
Plasma
M2 Macrophage
CD8+ T
Nerve
CD49a+ Smooth muscle
CD66+ Epithelial
TA
CD4+ T
CD36 low Endothelial
Goblet
CD36 high Endothelial
MUC1+ Epithelial
ITLN+ Epithelial
DC
PDPN+ Stromal
B
Cycling TA
M1 Macrophage
Paneth
Neuroendocrine
Neutrophil
ICC
CD57+ Epithelial
NK
C
D
Reference annotations
Leiden
FlowSOM
SpatialSort
PIXIE
UMAP 2
UMAP 1
E
Weighted F1
0.9
0.7
0.5
0.3
Leiden
FlowSOM
SpatialSort
PIXIE
G
Cluster purity
0.9
0.7
0.5
0.3
Leiden
FlowSOM
SpatialSort
PIXIE
F
F1
0.0
0.5
1.0
Smooth muscle
Endothelial
Plasma
M2 Macrophage
Enterocyte
Nerve
CD8+ T
Goblet
CD4+ T cell
Stroma
CD66+ Enterocyte
Neutrophil
Neuroendocrine
B
Cycling TA
Lymphatic
ICC
DC
MUC1+ Enterocyte
CD7+ Immune
Leiden
FlowSOM
SpatialSort
PIXIE
0
25k
Cells
H
Recall (%)
0
50
100
Smooth muscle
Endothelial
Plasma
M2 Macrophage
Enterocyte
Nerve
CD8+ T
Goblet
CD4+ T cell
Stroma
CD66+ Enterocyte
Neutrophil
Neuroendocrine
B
Cycling TA
Lymphatic
ICC
DC
MUC1+ Enterocyte
CD7+ Immune
Leiden
FlowSOM
SpatialSort
PIXIE
0
25k
Cells